

HOW CAN THE SEVERITY OF A ROAD CRASH BE USED TO DETERMINE THE ECONOMIC IMPACT

Motivation

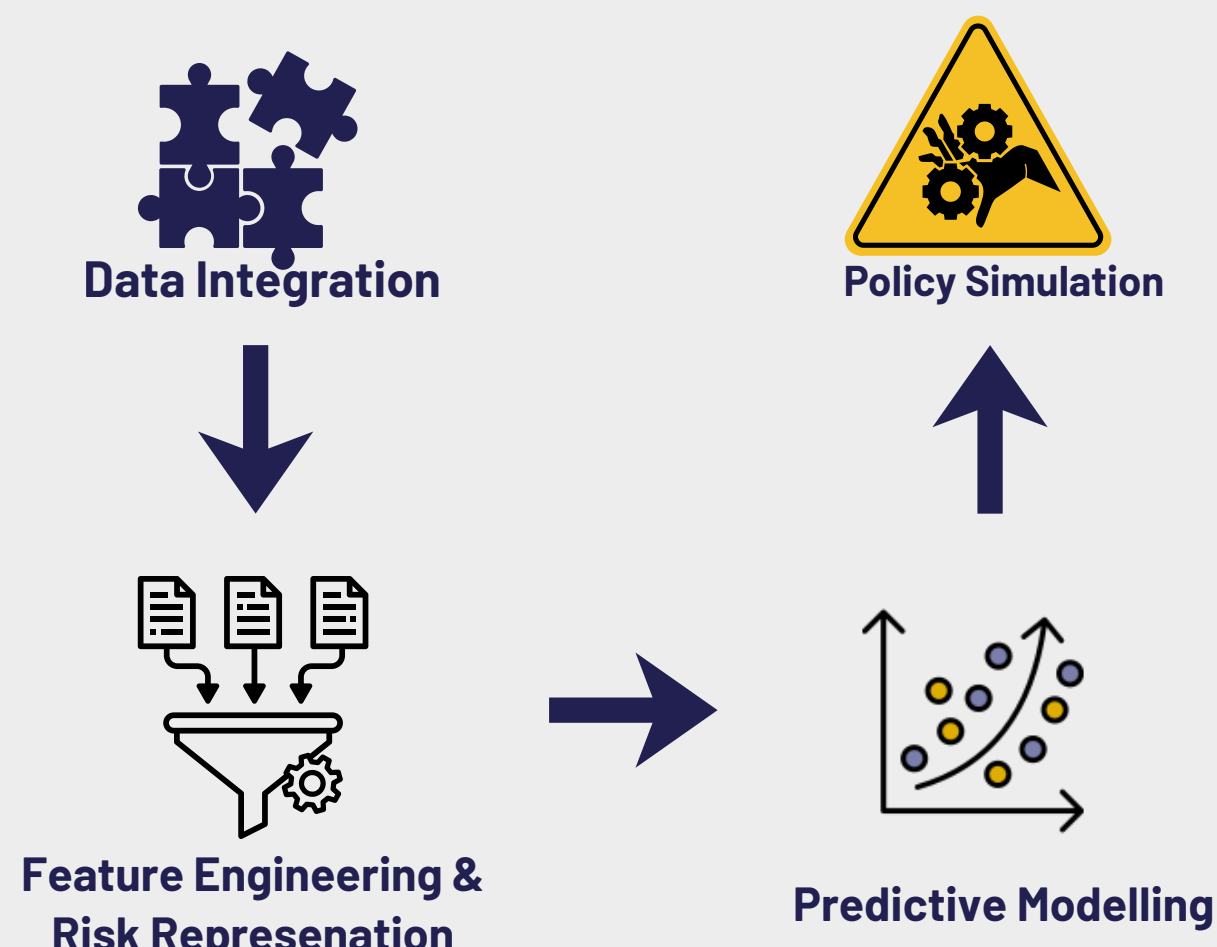
- Road traffic accidents remain a critical global public health challenge.
- Crash records alone describe what happened in the past, but they often do not fully capture the roadway environment that may contribute to pedestrian danger.
- The severity of a road crash determines its economic impact, using MAIS and VSL estimates from the DOT.

Research Questions

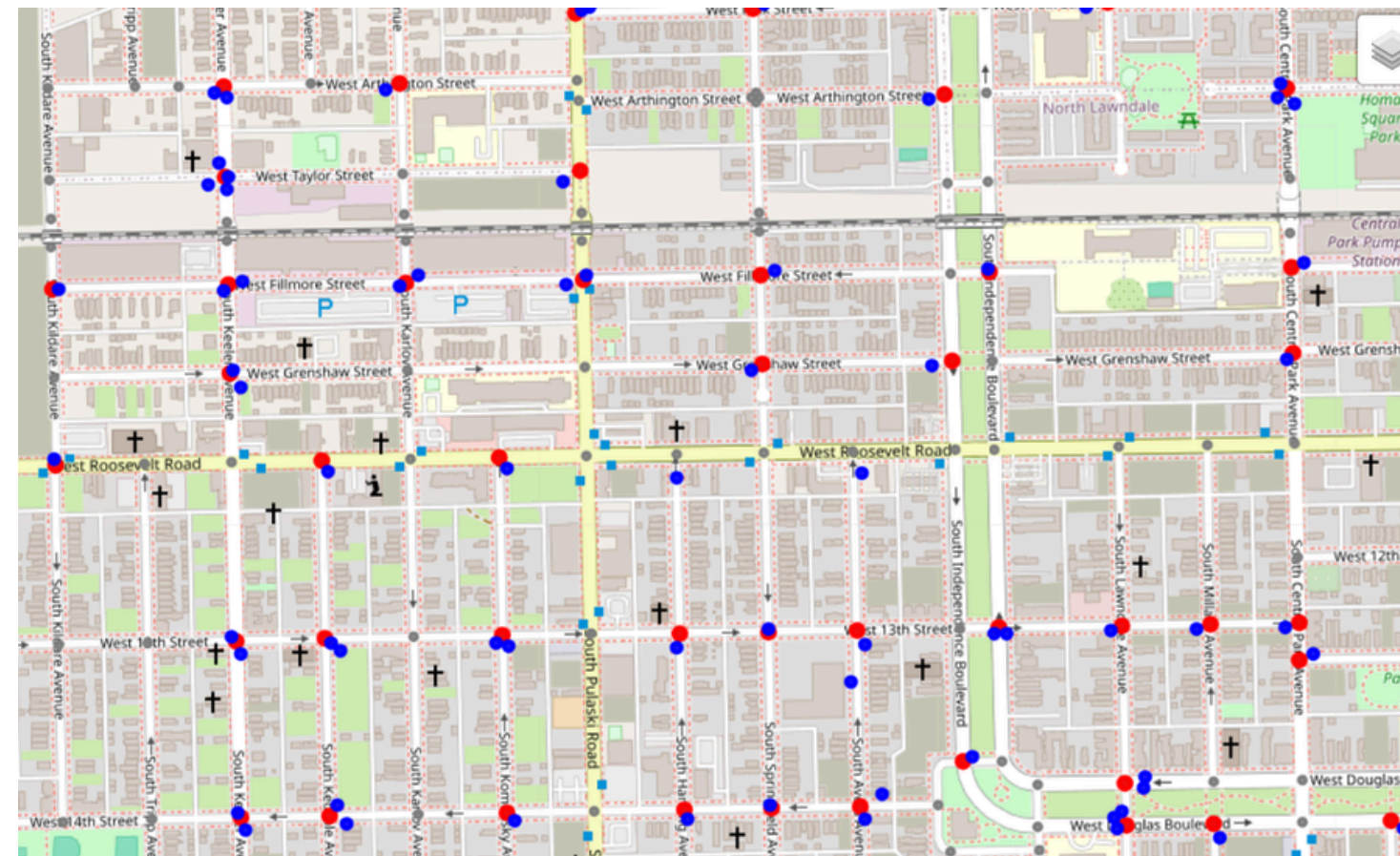
RQ1: Does traffic feature data improve accuracy when predicting injury severity for pedestrian accidents?

RQ2: How do traffic features reduce the injury severity and economic impact for pedestrian accidents?

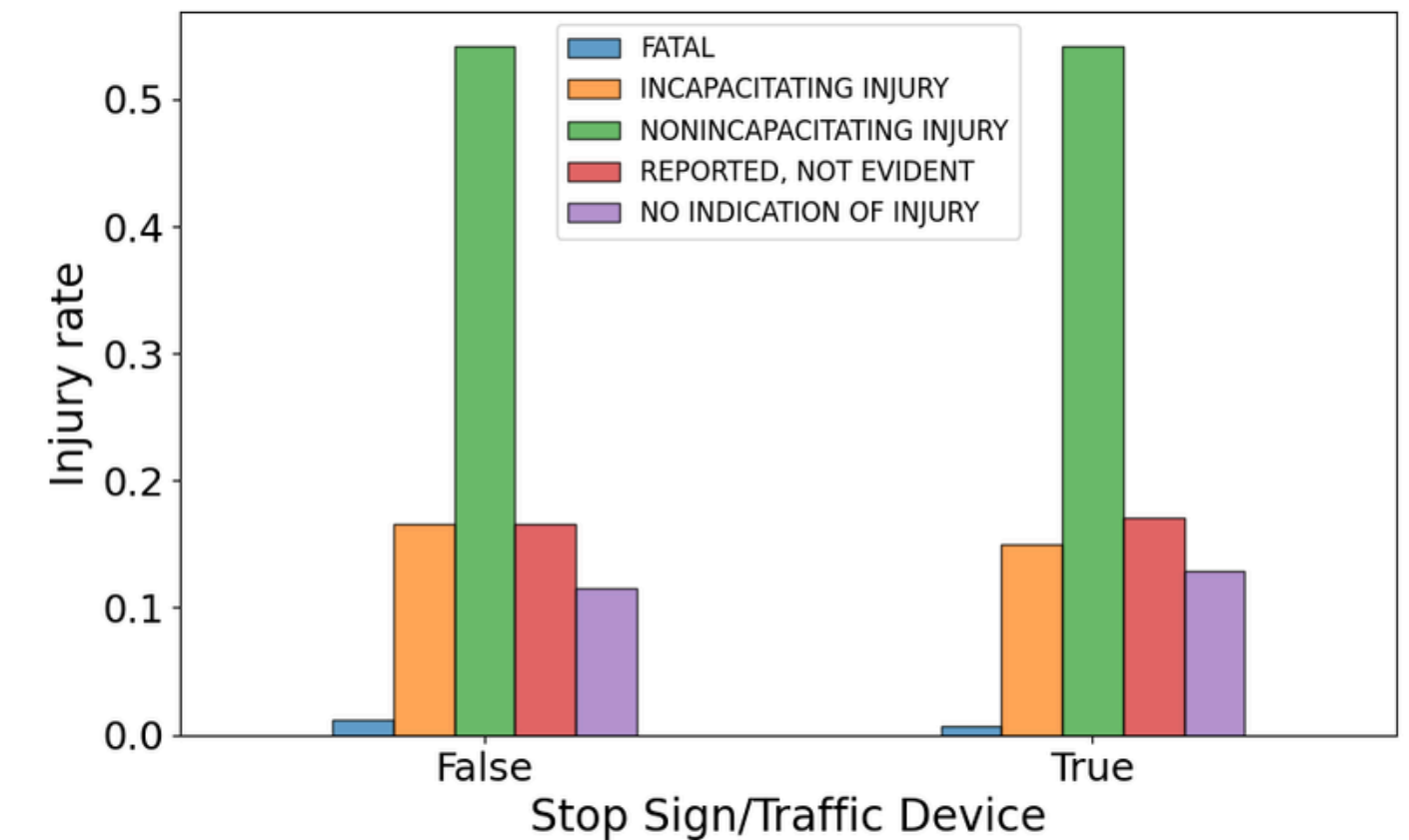
Methodology



Results



1. Road intersections (red) with stop signs (blue) using Mapillary API data



3. Injuries with and without a stop sign or traffic device

	Model	Accuracy	Precision	Recall	F1	ROC_AUC
0	Logistic Regression	0.621441	0.236559	0.580986	0.336220	0.645475
1	Logistic Regression, No Traffic Features	0.621732	0.235971	0.577465	0.335036	0.646013
2	Random Forest	0.783556	0.322645	0.283451	0.301781	0.655322
3	Random Forest, No Traffic Features	0.773388	0.313380	0.313380	0.313380	0.658011
4	DNN (Hidden Layers: (64, 32))	0.587740	0.232222	0.649648	0.342142	0.653175

2. Injury severity prediction performance for LR, RF, and DNN models

Discussion

By enriching crash history with street-level infrastructure information, our system not only predicts the injury severity of pedestrians in accidents, but also simulates how safety interventions could change that risk. Another aspect is calculating the economic impact of the crash by measuring its severity. Therefore, it could cost less to have a road sign or traffic device, than paying for the injury severity of the pedestrian following a road crash.

	Feature	n	Benefit per crash	CI 95 low	CI 95 high
0	Stop Sign/Traffic Device	6191	40543.35	37429.19	43657.50
1	Crosswalk	7231	2978.72	286.96	5670.49

4. RF predicted benefit (in dollars) for installing a stop sign/traffic device, or crosswalk

Conclusion

- RF beats LR and DNNs for predicting injury severity using crash accident report data, using raw accuracy.
- Adding traffic feature information collected from the Mapillary API offers slight increases to model performance, however more data may be necessary, such as the rate of speed of the vehicle.
- We use observational data and treatment assignment is not random, so interpretation should be based on model-based sensitivity results, not causal estimates.